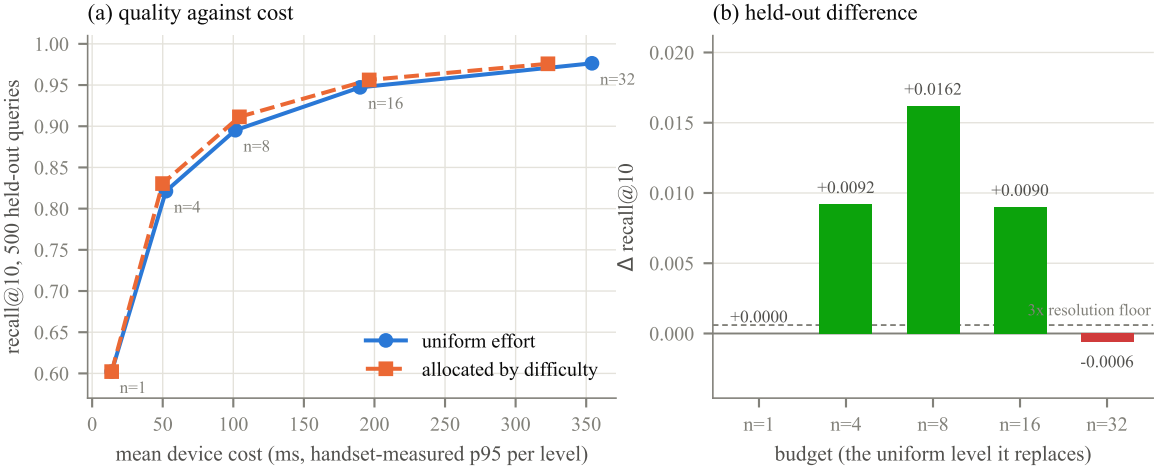
(a) quality against cost
recall@10, 500 held-out queries
mean device cost (ms, handset-measured p95 per level)
uniform effort
allocated by difficulty
n=1
n=4
n=8
n=16
n=32
(b) held-out difference
Δ recall@10
budget (the uniform level it replaces)
+0.0000
+0.0092
+0.0162
+0.0090
-0.0006
3× resolution floor